# Aluminum Control Dampers

| Item | Tag | Qty | Model | Dimension(s) | Airflow | Free Area Velocity | Free Area | Pressure Drop | Water Penetration | Sections |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | CD-100 | 22.75w x 49.75h | | | | | | 1 x 1 |

# Lifetime Fitness

**Quote: Q74332**
**Quote Date: 10-14-2026**

| | |
|---|---|
| **Company**<br>Engineered Air Solutions, LLC | **Branch**<br>Engineered Air Solutions, LLC - NJ |
| **Country**<br>United States of America | **State/Region**<br>New York |
| **Estimated Release Date**<br>12-2036 | **Valid through**<br>11-13-2026 |

## Aluminum Control Dampers

| Line # | Tag | Model | Qty |
|---|---|---|---|
| #1 | | CD-100 | 1 |

• Width: 23 • Height: 50 • Width Per Section: 22.75 • Height Per Section: 49.75 • Manufacturing Width: 22.75 • Manufacturing Height: 49.75 • Sizing: 1/4" Undersized

| Option |
|---|
| Frame and Blades |
| • Type: Extruded Aluminum ( .081 Frame and .125 Blades) |
| Operator |
| • Quantity Per Assembly : 1 • Position: In air stream • Operator Shaft Type: Direct Drive (No Jack Shaft) • Type: Electric Actuator • Operator Location: Right hand (from Jack Shaft) • Voltage: 24v (Modulating) • Operator Distribution: Fewest number of actuators • Operator: AFB-24-SR • Direction of Rotation: Power Open/Spring Closed |
| Blade Seal |
| • Type: TPV (175 Degrees Fahrenheit) |
| Jamb Seal |
| Bearings |
| • Type: Nylon |

United Enertech

(ENGINEERS)

# SUBMITTAL DATA

## CONTROL DAMPERS

Standard Blade

## OPPOSED BLADE - PARALLEL BLADE DAMPER

### Suggested Specifications:

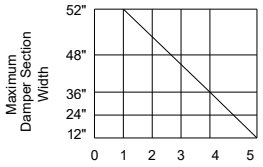
Furnish and install at location shown on drawing or in accordance with schedules dampers meeting the following specifications: Rectangular damper shall have 0.125" [3.18mm] extruded aluminum blades and .081" [2.06] extruded aluminum frame. CD-100, 101 meets the leakage requirements for Minimum AntiTerrorism Standards for Buildings (UFC 4-010-01, section B-4.2.3, max. leakage rates of 3 cfm/sq. ft. with differential pressure of 1" [25mm] w.g.). Blade and jamb seals must be used to achieve optimal leakage rating. Damper to have thrust bushings and meet the low pressure drop equal to United Enertech MODEL CD-100 or CD-101.

FPM Table

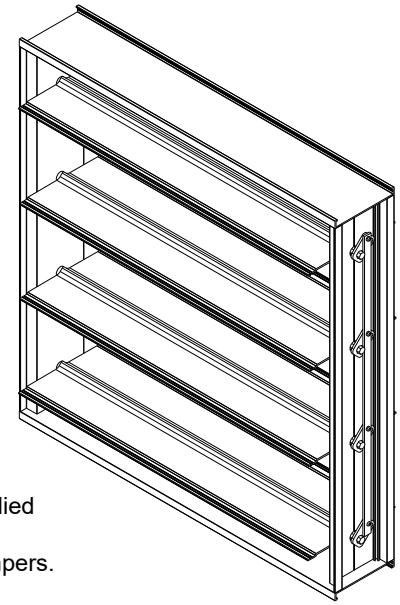
| | |
|---|---|
| 12" wide | - 2600 FPM |
| 24" | - 2200 |
| 36" | - 2000 |
| 48" | - 1700 |
| 52" | - 1600 |

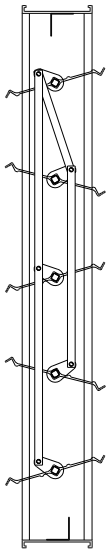
Pressure - up to 3" [76mm] w.g. - See pressure limit table

Pressure Limitations




Jackshafting provided over 52" [1321mm] wide





### Standard Construction:

**Frame:** 0.081 [2.06mm] Extruded Aluminum

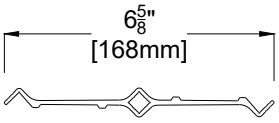
**Blade:** 0.125 [3.18mm] Extruded Aluminum

**Bearing:** Nylon

**Linkage:** Concealed in frame

**Axles:** 3/8" [9.35mm] square, zinc plated steel

**Control Shaft:** $\frac{1}{2}$" x 6" [13mm x 152mm] long outboard support bearing supplied with all single section dampers for field mounted actuators. Factory-installed jackshaft supplied with all multi-section dampers.




Blade Detail

Undersized 1/4" [6.35mm]

Minimum Size: 5"w x 4"h [127mm x 102mm]
8" [203mm] and under single blade

Maximum Size: 52"w x 72"h [1320mm x 1829mm]
(single section) Multi-section: unlimited

### Options:

- ❑ 0.125 [3.18mm] Extruded Aluminum Frame (box frame)
- ❑ 0.125 [3.18mm] Extruded Aluminum Frame w/ 1-1/2" [38mm] integral flange
- ❑ Bronze Bushing
- ❑ Blade Seals PVC (175°F [79°C])
- ❑ Blade Seals Silicone (400°F [204°C] & #304 Stainless Steel Bushing)
- ❑ Compression Jamb Seals (stainless steel)
- ❑ Header Plates (End Flange)
- ❑ Hand Quadrant
- ❑ Factory Installed Pneumatic or Electric Actuators (see cat. sheet H-1)
- ❑ Stand Off Bracket, 2" [51mm]
- ❑ Position Switch
- ❑ C02 sensor (24V)
  - ❑ 0-5, 0-10 vdc output (without display)
  - ❑ 0-5, 0-10 vdc output (with display)
  - ❑ 4-20 mA output (without display)
- ❑ Face and By-pass Damper
- ❑ Chain Operator
- ❑ Heresite coated (air dry)
- ❑ Epoxy coated (powder coated @ 415°[213° C])
- ❑ Weather Shield Enclosure for Actuator
- ❑ Explosion-proof Enclosure for Actuator

INDEPENDENT LABORATORY CERTIFIED air leakage AMCA® standard 500 Certaire Technical Services, LTD.

size tested: 36"w x 36"h [914mm x 914mm] 12"w x 12"h [305mm x 305mm]

INDEPENDENT LABORATORY CERTIFIED air performance AMCA® standard 500 Certaire Technical Services, LTD.

size tested: 36"w x 36"h [914mm x 914mm] 12"w x 12"h [305mm x 305mm]



Due to continuing research, United Enertech reserves the right to change specifications without notice.

| | |
|---|---|
| Job Name: | ☐ MODEL CD-100 (Opposed) |
| Location: | ☐ MODEL CD-101 (Parallel) |
| Architect: | |
| Engineer: | |
| Contractor: | |

| | | |
|---|---|---|
| DRAWN BY: CLJ | DATE: 11-20-01 | REV. DATE: 12/1/21 |
| REV. NO. 48 | APPROVED BY: CLJ | DWG. NO.: A-1 |

# Model CD-100, CD-101 Damper Performance Data

## LEAKAGE PERFORMANCE

United Enertech certifies that the CD-100-I, CD-101-I is licensed to bear the Certaire Certified Air Ratings Seal. The ratings shown are based on tests made in accordance with AMCA standard 500.

| Damper Width X Height | 1 in. w.g. Class | 4 in. w.g. Class | 8 in. wg Class |
|---|---|---|---|
| 12" x 12" | 3.6 (Class 1A) | 7.6 (Class 1A) | 10.3 (Class 1A) |
| 36" x 36" | 0.5 (Class 1A) | 5.6 (Class 1A) | 20.8 (Class 1A) |

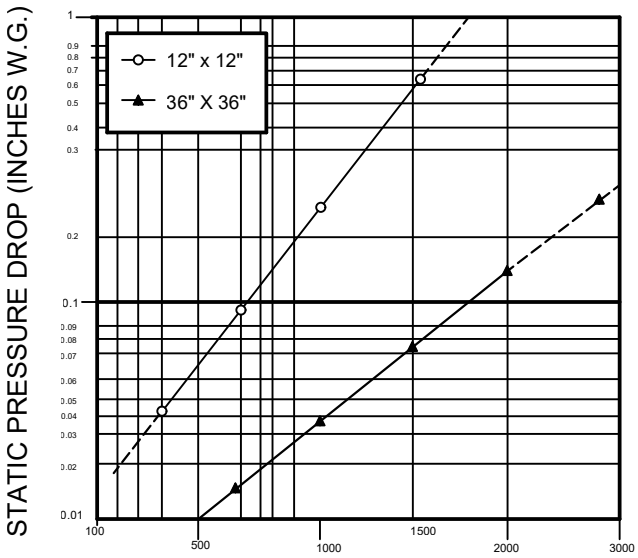
Note: At 1" w.g. 12"x12" damper leaks 3.6 cfm per square foot.
Note: At 1" w.g. 36"x36" damper leaks 0.5 cfm per square foot.

## AMCA STANDARDS

| Pressure / Class | Leakage, $ft^3/min/ft^2$ | | | |
|---|---|---|---|---|
| | Required Rating | | Extended Ranges (optional) | |
| | 0.25 kPa (1") | 1.0 kPa (4") | 2.0 kPa (8") | 3.0 kPa (12") |
| 1A | 15.2 (3) | 40.6 (8) | 55.9 (11) | 71.1 (14) |
| 1 | 20.3 (3) | 40.6 (8) | 55.9 (11) | 71.1 (14) |
| 2 | 50.8 (10) | 102 (20) | 142 (28) | 178 (35) |
| 3 | 203 (40) | 406 (80) | 569 (112) | 711 (140) |

Leakage Classification per AMCA publication 511-99

## PRESSURE DROP




Face Velocity (FT/MIN)

Based on STANDARD AIR- .075 lb. per cubic foot.

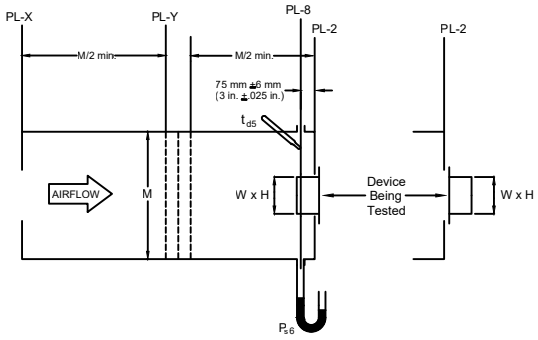



Figure 5.5- Test Device Setup with Inlet Chamber

12" x 12" (305mm x 305mm)

| Face Velocity ft/min (m/s) | Pressure drop w.g. (PA) |
|---|---|
| 407 (2.0) | .040 (1.24) |
| 820 (4.1) | .150 (4.97) |
| 1234 (6.3) | .340 (12.42) |
| 1641 (8.3) | .610 (22.35) |
| 2057 (10.4) | .950 (34.77) |

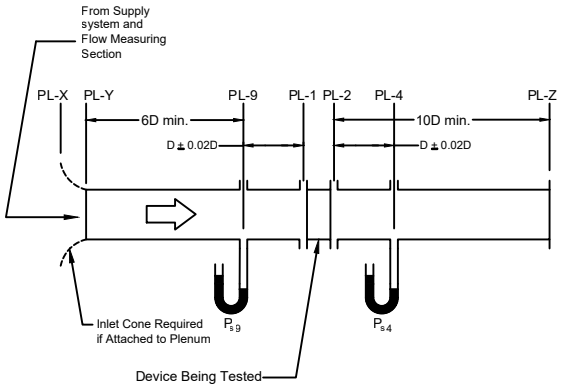



Figure 5.3- Test Device Setup with Inlet and Outlet Ducts

36" x 36" (914mm x 914mm)

| Face Velocity ft/min (m/s) | Pressure drop w.g. (PA) |
|---|---|
| 392 (1.99) | .005 (1.24) |
| 789 (4.01) | .02 (4.97) |
| 1208 (6.14) | .05 (12.42) |
| 1598 (8.12) | .09 (22.35) |
| 2006 (10.19) | .14 (34.77) |